Dax L. Feliz, PhD | Curriculum Vitae

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Flatiron Institute
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Education

PhD Astrophysics, Vanderbilt University (VU), May 2022. Advisor: Dr. Keivan Stassun, Dissertation Link

M.A. Physics, Fisk University, May 2018. Advisors: Drs. Keivan Stassun and Karen Collins

B.S. Astrophysics, University of Massachusetts at Amherst, May 2013. Advisor: Dr. Min Yun

Academic Positions

Flatiron Research Fellow, Flatiron Center for Computational Astrophysics (CCA), September 2024 – Present
Postdoctoral Advisor: Dr. David Hogg

Postdoctoral Researcher, American Museum of Natural History (AMNH), September 2022 – September 2024
Postdoctoral Advisor: Dr. Ruth Angus

Postdoctoral Researcher, Vanderbilt University, May 2022 - August 2022
Postdoctoral Advisor: Dr. Keivan Stassun

Research Interests

Detection and characterization of exoplanets orbiting cool M-dwarfs.

End-to-end transit-photometry pipelines for ground- and space-based surveys (KELT, Kepler, TESS, PLATO).

Stellar activity in time-series photometry: rotation, flare statistics, and activity cycles.

Research Experience

Transit Survey of M-Dwarf stars in TESS data, VU/AMNH/CCA 2018 – Present

PhD Thesis: Built NEMESIS pipeline to extract TESS Full Frame Image light curves, remove instrumental systematics, and run periodic transit searches to surface high-value candidates.

PLATO Flare Removal Working Group, European Space Agency, 2020 – Present

Constructing an algorithm to detect and remove flare events from the photometric time series data that will be produced by the ESA PLATO mission.

Kilodegree Extremely Little Telescope (KELT), VU, TN 2017 – 2022

Operating KELT data reduction pipeline for detection of transiting exoplanets orbiting bright, hot stars with a network of ground-based telescopes.

Multi-Year Transit Survey of Proxima Centauri, Fisk University and VU, TN 2016 – 2019

Master's Thesis Project: Processing images from observations of the star Proxima Centauri, production and detrending analysis of light curves and utilization of periodic transit search algorithms to hunt for transit events of the Earth-sized planet, Proxima b.

Light Curve Analysis of high proper motion M-Dwarfs in Kepler data, AMNH, NY 2012

REU Summer Project: Analysis and signal processing of photometry from the Kepler Space Telescope, conducted a search for transiting exoplanets and gravitational microlensing events. Advisor: Dr. Sebastine Lépine.

Using Spectropolarimetry to Study Extrasolar Planets, AMNH, NY 2010

REU Summer Project: A theoretical project where I sought to answer if exoplanet atmospheres could be detected and characterized from ground-based telescopes. Advisors: Drs. Rebecca Oppenheimer & Douglas Brenner.

Mentoring Experience

Student Project Totals by Category: High School Students: 7 ; College Students: 5 ; Graduate Students: 2.

CUNY Master's in Astrophysics, Bridge Thesis Project
Celestial Fireworks: Using TESS Photometry and SDSS Spectra to Uncover Stellar Eruptions, 2024 – *Present*. Graduate Student: Andrea Bracamonte, City University of New York Graduate Center, Co-Advised with Dr. Ruth Angus.

Barnard Summer Research Institute AMNH Project
Detection and Validation of Transiting Exoplanets Around Nearby M-dwarf Stars, May 2024 - August 2024. Graduate Student: Madeline J. Maldonado Gutierrez, Barnard College of Columbia University.

CUNY Master's in Astrophysics, Bridge Thesis Project
Using Gaussian Processes to model stellar rotation in Zwicky Transient Facility photometry, 2022 - 2024. Graduate Student: Ryan Lebron, City University of New York Graduate Center, Co-Advised with Dr. Ruth Angus.

AMNH Research Experience for Undergraduates (REU) Program
Detecting Exoplanets Around Nearby M-dwarf Stars from the Revised TESS Habitable Zone Catalog, 2023. Undergraduate Student: Maliyah Adams, Arizona State University

AMNH Science Research Mentoring Program
Detection of Transiting Exoplanets and Eclipsing Binaries Around Nearby M-dwarf Stars, 2022 - 2024. High School Students: August Fischer, Donovan Bradley, Jashcelyn Canada (2022 – 2023), Kylor Ghai, Shreeya KC, Thamim Chowdhury (2023-2024)

Central American-Caribbean Bridge Program
Project: Recovery of known TESS Objects of Interest from Full Frame Images of TESS observations, 2020 - 2021. Undergraduate Student: Bryan Villarreal Alvarado, University of Costa Rica.

Summer Research Internship
Project: Transiting Exoplanet Survey Satellite: A Search for New Worlds, 2019 - 2020.
Undergraduate Student: Samantha Bianco, Vanderbilt University.

Summer Research Internship
Project: Blind Transit Survey of TESS Data for M Dwarf Systems, 2019 - 2020.
Undergraduate Student: Mary Jimenez, George Mason University.

School for Science and Math at Vanderbilt
Project: Detecting exoplanet transits in TESS light curves, Spring 2019.
High School Student: Felix Bean, Hunter's Lane High School.

East Harlem Tutorial Program, New York, NY, 2008 – 2010.
Tutored and mentored many high-school and middle school students in math, science and the college application process. I was also a mentor in the For Inspiration and Recognition of Science and Technology (FIRST) robotics competition.

Teaching Experience

CUNY Master's in Astrophysics Bridge Program
Programming Bootcamp, August 2022, 2023, 2024 & 2025.
A two week orientation and programming bootcamp designed to teach new astronomy graduate students of various backgrounds a myriad of skills they may require in to begin their academic journeys as well as integrate them in the NYC astronomy community.

Teaching Assistant

ASTR-101L: Introductory Solar System Laboratory, Vanderbilt University, TN Fall semesters, 2018 – 2020.

ASTR-102L: Introductory Stars & Galaxies Laboratory, Vanderbilt University, TN Spring semesters, 2018 – Spring 2020.

Laboratory Technician / Teaching Assistant

Electronics Laboratory, Bronx Community College, NY Spring 2014 – Fall,2015.

Outreach

East Harlem Scholars Academy, New York, NY, Alumni Day Speaker, 2010 – 2015
- Answering questions of elementary school students about astronomy and physics and my research experience.

Central Park East High School, New York, NY, College Panel Speaker, 2009 – 2013
- Answering questions of high-school students about my experiences in science and in higher education.

Central Park East High School, New York, NY, Robotics Consultant, 2009 – 2011
- Volunteered to assist high school students in the FIRST Robotics Competition.

Awards, Grants, Activities & Honors

Visiting Graduate Student Fellow at George Mason University, 2021 – 2022.

NSF AGEP Fellowship, Spring 2021 – 2022.

Fisk-Vanderbilt Bridge Fellowship, Fall 2018 – 2021.

McMinn Summer Research Award, 2018.

Graduate Opportunities at Fisk in Astronomy and Astrophysics Research (GO-FAAR) grant, 2016 – 2018.

Heroes Classroom Namesake, East Harlem Scholars Academy, New York, NY, 2013

New York Space Consortium Research Grant, Summer of 2012.

Invited Talks, Presentations, and Posters

"NEMESIS II: Exoplanet Transit Survey of Nearby M-dwarfs in TESS FFIs" — delivered at: TESS Science Conference III (Jul 2024; poster; `https://zenodo.org/records/12973447`), Princeton EREX X (Jun 2025; talk), Caltech Sagan Summer Workshop (Jul 2025; poster `https://nexsci.caltech.edu/workshop/2025/posters/Poster_DaxFeliz_101.pdf`), University of New Mexico "From Trends to Transits" Workshop (Aug 2025; talk).

"Can Photometric Flares Detected by Ground and Space Based Telescopes Be Used To Identify Activity Cycles in M-dwarf stars?", Cool Stars 22 (Jul 2024; poster `https://zenodo.org/records/13146889`)
"Twinkle, Twinkle Little Star", Twinkle and the Next Generation of Exoplanet Scientists Conference (Sep 2021).

"NEMESIS I: Exoplanet Transit Survey of Nearby M-dwarfs in TESS FFIS I": Presented at MIT TESS Science Talk (Mar 2021); Rubin LSST TVS Seminar (Jun 2021); TESS Science Conference II (Aug 2021; `http://doi.org/10.5281/zenodo.5115302`); Center for Exoplanets & Habitable Worlds Seminar, Penn State (Sep 2021); Cool Stars 20.5 (Mar 2021; poster `http://doi.org/10.5281/zenodo.4562794`).

"A Multi-Year Transit Search of Proxima Centauri", Master's Thesis Defense, Fisk University (Aug 2018); ERES III, Yale (Jun 2017); Sagan Summer Workshop, Caltech (Aug 2017).

"Using BLS modeling on photometry of Proxima Centauri", KELT Workshop, Lehigh University (Jun 2017).

"How to deflect an incoming asteroid", University of Massachusetts at Amherst Astronomy Colloquium, (May 2013).

"Light Curve Characterization of High Proper Motion M-Dwarf Stars with the Kepler Space Telescope", AMNH REU Annual Symposium (Aug 2012).

"Using Spectropolarimetry to Study Extrasolar Planets", AMNH REU Annual Symposium (Aug 2010)

Publication Metrics

Refereed: 31 / 1st Author: 3 / Citations: 855 / h-index: 16 (2025-10-18, via ADS).

Peer Reviewed Publications (1st Author)

1. **Dax L. Feliz**, Peter Plavchan, Samantha N. Bianco, Mary Jimenez, Kevin I. Collins, Bryan Villarreal Alvarado, and Keivan G. Stassun. NEMESIS: Exoplanet Transit Survey of Nearby M-dwarfs in TESS FFIs. I. *The Astronomical Journal*, 161(5):247, May 2021

2. **Dax L. Feliz** L., David L. Blank, Karen A. Collins, and ... A Multi-year Search for Transits of Proxima Centauri. II. No Evidence for Transit Events with Periods between 1 and 30 days. *The Astronomical Journal*, 157(6):226, Jun 2019

3. David L. Blank, **Dax L. Feliz** , Karen A. Collins, and ... A Multi-year Search for Transits of Proxima Centauri. I. Light Curves Corresponding to Published Ephemerides. *The Astronomical Journal*, 155(6):228, Jun 2018** **Corresponding author**

Peer Reviewed Publications (Nth Author)

1. Heike Rauer, Conny Aerts, Cabrera, ..., **D. L. Feliz**, and ... The PLATO mission. *Experimental Astronomy*, 59(3):26, June 2025

2. N. Heidari, G. Hébrard, E. Martioli, ..., **D. L. Feliz**, and ... Characterization of seven transiting systems, including four warm Jupiters from SOPHIE and TESS. *Astronomy & Astrophysics*, 694:A36, February 2025

3. E. Semenko, O. Kochukhov, Z. Mikulášek, ..., **D. L. Feliz**, and ... HD 34736: an intensely magnetised double-lined spectroscopic binary with rapidly rotating chemically peculiar B-type components. *Monthly Notices of the Royal Astronomical Society*, 535(3):2812–2836, December 2024

4. Rishi R. Paudel, Thomas Barclay, Allison Youngblood, ..., **Dax L. Feliz**, and ... A Multiwavelength Survey of Nearby M Dwarfs: Optical and Near-ultraviolet Flares and Activity with Contemporaneous TESS, Kepler/K2, Swift, and HST Observations. *The Astrophysical Journal*, 971(1):24, August 2024

5. Lionel J. Garcia, Daniel Foreman-Mackey, Catriona A. Murray, ..., **Dax L. Feliz**, and ... nuance: Efficient Detection of Planets Transiting Active Stars. *The Astronomical Journal*, 167(6):284, June 2024

6. Christopher R. Mann, Paul A. Dalba, David Lafrenière, ..., **Dax L. Feliz**, and ... Giant Outer Transiting Exoplanet Mass (GOT 'EM) Survey. III. Recovery and Confirmation of a Temperate, Mildly Eccentric, Single-transit Jupiter Orbiting TOI-2010. *The Astronomical Journal*, 166(6):239, December 2023

7. Justin M. Wittrock, Peter P. Plavchan, Bryson L. Cale, ..., **Dax L. Feliz**, and ... Validating AU Microscopii d with Transit Timing Variations. *The Astronomical Journal*, 166(6):232, December 2023

8. Ismael Mireles, Diana Dragomir, Hugh P. Osborn, ..., **Dax L. Feliz**, and ... TOI-4600 b and c: Two Long-period Giant Planets Orbiting an Early K Dwarf. *The Astrophysical Journal Letters*, 954(1):L15, September 2023

9. Keivan G. Stassun, Guillermo Torres, Marina Kounkel, ..., **Dax L. Feliz**, and ... An Eclipsing Binary Comprising Two Active Red Stragglers of Identical Mass and Synchronized Rotation: A Post-mass-transfer System or Just Born That Way? *The Astrophysical Journal*, 950(2):99, June 2023

10. Joseph E. Rodriguez, Samuel N. Quinn, Andrew Vanderburg, ..., **Dax L. Feliz**, and ... Another Shipment of Six Short-Period Giant Planets from TESS. *Monthly Notices of the Royal Astronomical Society*, February 2023

11. Mohammed El Mufti, Peter P. Plavchan, Howard Isaacson, ..., **Dax L. Feliz**, and ... TOI 560: Two Transiting Planets Orbiting a K Dwarf Validated with iSHELL, PFS, and HIRES RVs. *The Astronomical Journal*, 165(1):10, January 2023

12. Keivan G. Stassun, Guillermo Torres, ..., **Dax L. Feliz**, and ... A Low-mass Pre-main-sequence Eclipsing Binary in Lower Centaurus Crux Discovered with TESS. *The Astrophysical Journal*, 941(2):125, December 2022

13. S. Ulmer-Moll, M. Lendl, S. Gill, ..., **D. L. Feliz**, and ... Two long-period transiting exoplanets on eccentric orbits: NGTS-20 b (TOI-5152 b) and TOI-5153 b. *arXiv e-prints*, page arXiv:2207.03911, July 2022

14. Ian Stotesbury, Billy Edwards, ..., **Dax L. Feliz**, and ... Twinkle: a small satellite spectroscopy mission for the next phase of exoplanet science. In Laura E. Coyle, Shuji Matsuura, and Marshall D. Perrin, editors, *Space Telescopes and Instrumentation 2022: Optical, Infrared, and Millimeter Wave*, volume 12180 of *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, page 1218033, August 2022

15. Justin M. Wittrock, Stefan Dreizler, Michael A. Reefe, Brett M. Morris, ..., **D. L. Feliz**, and ... Transit Timing Variations for AU Microscopii b and c. *The Astronomical Journal*, 164(1):27, July 2022

16. Michael A. Reefe, Rafael Luque, Eric Gaidos, ..., **D. L. Feliz**, and ... A Close-in Puffy Neptune with Hidden Friends: The Enigma of TOI 620. *The Astronomical Journal*, 163(6):269, June 2022

17. Jiayin Dong, Chelsea X. Huang, George Zhou, ..., **D. L. Feliz**, and ... NEID Rossiter-McLaughlin Measurement of TOI-1268b: A Young Warm Saturn Aligned with Its Cool Host Star. *The Astrophysical Journal Letters*, 926(2):L7, February 2022

18. Christina Hedges, Alex Hughes, George Zhou, ..., **D. L. Feliz**, and ... TOI-2076 and TOI-1807: Two Young, Comoving Planetary Systems within 50 pc Identified by TESS that are Ideal Candidates for Further Follow Up. *The Astronomical Journal*, 162(2):54, August 2021

19. Keivan G. Stassun, Guillermo Torres, Cole Johnston, ..., **Dax L. Feliz**, and ... Discovery and Characterization of a Rare Magnetic Hybrid β Cephei Slowly Pulsating B-type Star in an Eclipsing Binary in the Young Open Cluster NGC 6193. *The Astrophysical Journal*, 910(2):133, April 2021

20. S. A. Rappaport, D. W. Kurtz, G. Handler, ..., **D. L. Feliz**, and ... A tidally tilted sectoral dipole pulsation mode in the eclipsing binary TIC 63328020. *Monthly Notices of the Royal Astronomical Society*, February 2021

21. Joseph E. Rodriguez, Samuel N. Quinn, George Zhou, ..., **Dax L. Feliz**, and ... TESS Delivers Five New Hot Giant Planets Orbiting Bright Stars from the Full-frame Images. *The Astronomical Journal*, 161(4):194, April 2021

22. Joni-Marie C. Cunningham, **Dax L. Feliz**, Don M. Dixon, and ... A KELT-TESS Eclipsing Binary in a Young Triple System Associated with the Local "Stellar String" Theia 301. *The Astrophysical Journal*, 160(4):187, October 2020

23. Peter Plavchan, Thomas Barclay, Jonathan Gagné, ... **Dax Feliz** , and ... A planet within the debris disk around the pre-main-sequence star AU Microscopii. ***Nature***, 582(7813):497–500, June 2020

24. Romy Rodríguez Martínez, B. Scott Gaudi, Joseph E. Rodriguez, ..., **Dax L. Feliz** , and ... KELT-25 b and KELT-26 b: A Hot Jupiter and a Substellar Companion Transiting Young A Stars Observed by TESS. *The Astronomical Journal*, 160(3):111, September 2020

25. Weicheng Zang, Subo Dong, Andrew Gould, . .., **Dax L. Feliz**, and . .. Spitzer + VLTI-GRAVITY Measure the Lens Mass of a Nearby Microlensing Event. *The Astrophysical Journal*, 897(2):180, July 2020

26. Joseph E. Rodriguez, Jason D. Eastman, George Zhou, ..., **Dax L. Feliz** , and ... KELT-24b: A 5M_J Planet on a 5.6 day Well-aligned Orbit around the Young V = 8.3 F-star HD 93148. *The Astronomical Journal*, 158(5):197, Nov 2019

27. Jonathan Labadie-Bartz, Joseph E. Rodriguez, Keivan G. Stassun, ..., **Dax L. Feliz** , and ... KELT-22Ab: A Massive, Short-Period Hot Jupiter Transiting a Near-solar Twin. *The Astrophysical Journal*, 240(1):13, Jan 2019

28. Karen A. Collins, Kevin I. Collins, Joshua Pepper, ..., **Dax L. Feliz** , and ... The KELT Follow-up Network and Transit False-positive Catalog: Pre-vetted False Positives for TESS. *The Astronomical Journal*, 156(5):234, Nov 2018